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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 11	
TITLE : Interface Implementation	

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Exercise

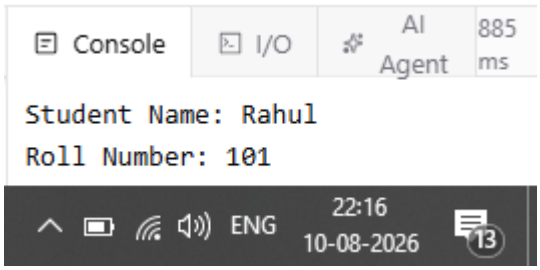
1. Create an interface Printable and implement it in Student and Employee classes.

Code:

```
interface Printable {
void printDetails();
}
class Student implements Printable {
String name;
int rollNo;
Student(String name, int rollNo) {
this.name = name;
this.rollNo = rollNo;
}
public void printDetails() {
System.out.println("Student Name: " + name);
System.out.println("Roll Number: " + rollNo);
}
}
class Employee implements Printable {
String name;
int employeeId;
Employee(String name, int employeeId) {
this.name = name;
this.employeeId = employeeId;
```

```
}
public void printDetails() {
System.out.println("Employee Name: " + name);
System.out.println("Employee ID: " + employeeId);
}
}
public class Main {
public static void main(String[] args) {
Student s = new Student("Rahul", 101);
Employee e = new Employee("Amit", 201);
s.printDetails();
System.out.println();
e.printDetails();
}
}
```

Output:



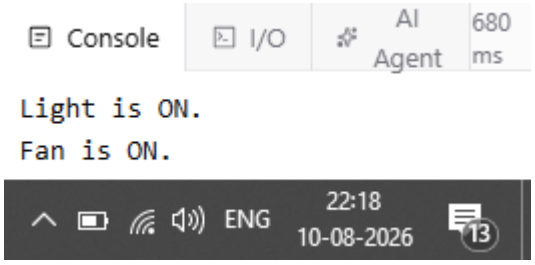
2.Create an interface Switchable with a method turnOn(). Implement the interface in Light and Fan classes to display the device status.

Code:

```
interface Switchable {
void turnOn();
}
class Light implements Switchable {
public void turnOn() {
System.out.println("Light is ON");
}
}
class Fan implements Switchable {
public void turnOn() {
System.out.println("Fan is ON");
}
}
public class InterfaceDemo {
public static void main(String[] args) {
```

```
Light l = new Light();
Fan f = new Fan();
l.turnOn();
f.turnOn();
}
}
```

Output:



Conclusion

Interfaces in Java are used to achieve abstraction and provide a common structure for different classes. A class implements an interface and provides its own definition for the methods. In this program, different classes implement the same interface and perform their respective operations. Thus, interfaces improve code reusability, flexibility, and maintainability in Java programs.